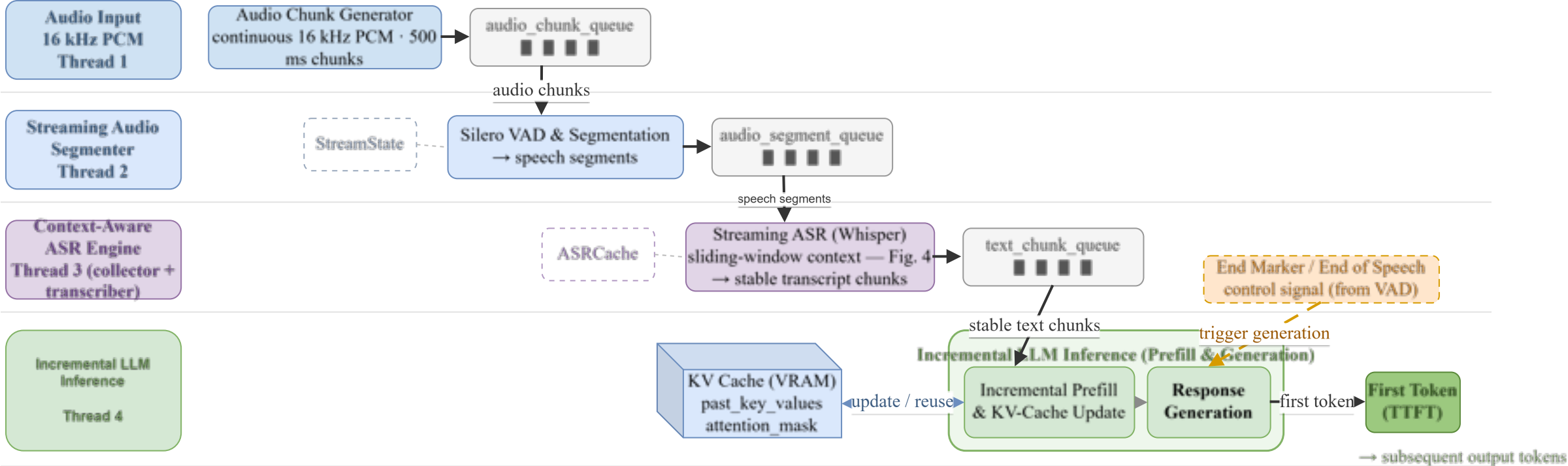
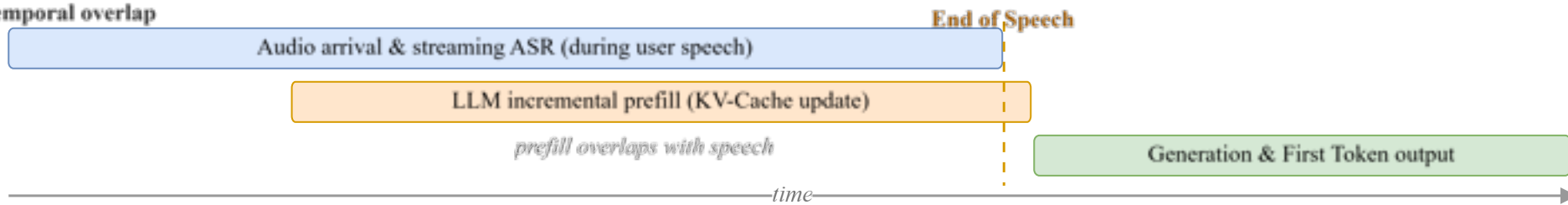




Pipeline parallelism — temporal overlap



Producer–consumer queues let ASR transcription and LLM prefill overlap with the user’s speech; after End-of-Speech only the last chunk and thread synchronization remain on the critical path.